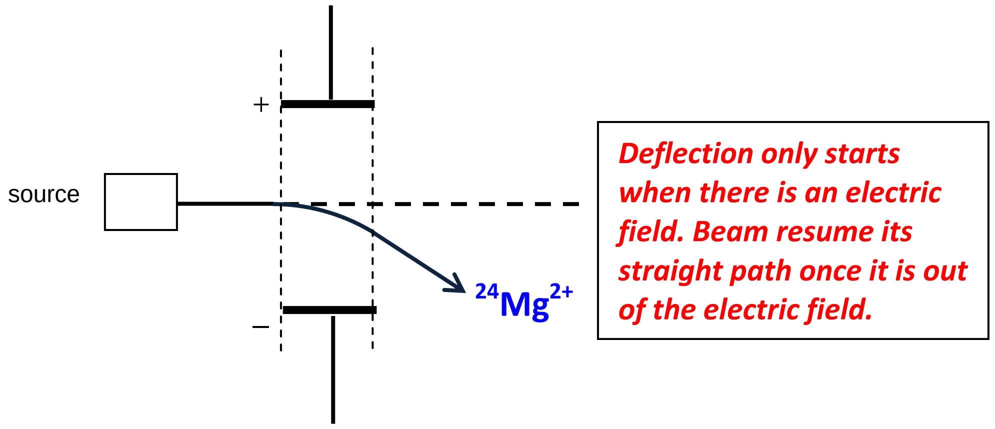
Suggested Mark Scheme for 2013 H1 Chemistry Prelim Paper 2

1. **(a)** **(i)**



(ii) Angle of deflection = $\frac{2.5}{2/24} \times \frac{2}{14}$ = **4.29 °**

(b) **(i)** **Lattice energy of Mg*Cl*$_2$**

(ii) Δ**H$_3$** = –1920 + 2(–381) = **–2682 kJ mol^{-1}**

Δ**H$_1$** = $\Delta H_3 - \Delta H_2$ = –(–2326)+ (–2682) = **–356 kJ mol^{-1}**

(iii) **Lattice energy MgO is more exothermic than that of Mg*Cl*$_2$** or **magnitude of the lattice energy of MgO is greater.**

This is because O^{2-} is smaller ionic radius and has higher charge than Cl^-.

2. **(a)** Methanoic acid is used in large excess so as to **keep its concentration** relatively **constant** throughout the experiment so that we **can determine the order of reaction with respect to Br_2** and the **rate is independent of concentration of HCOOH**.

(b) **$(t_{1/2})_1$= 175 s, $(t_{1/2})_2$= 180 s**

Average $(t_{1/2})_1$= 177.5 s

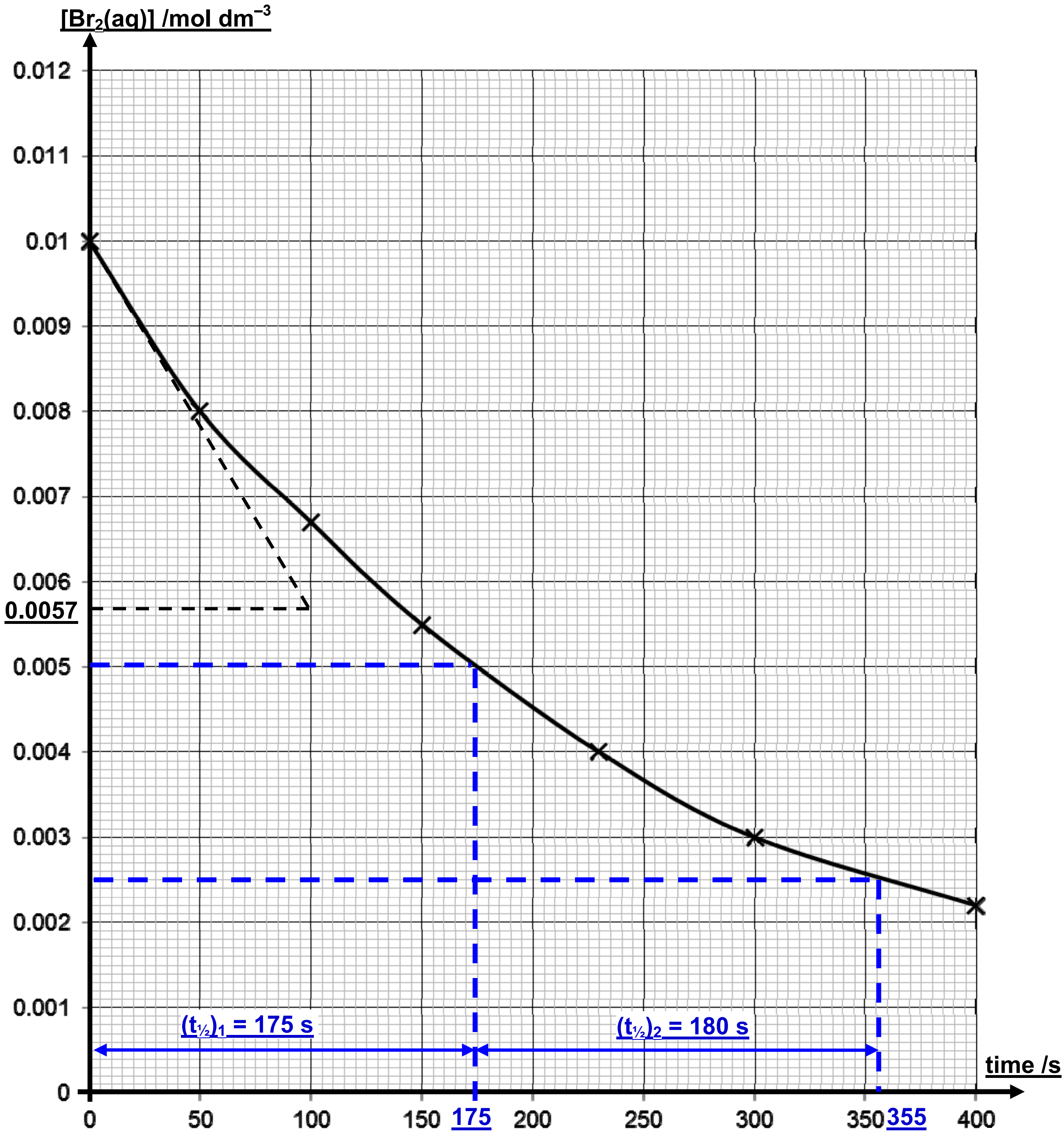
Successive half-life approximately constant, the reaction is first order with respect to Br_2.

(c) **Rate = k[methanoic acid][Br_2]**

(d) Initial rate = $|gradient|$ = $\left|\frac{0.010-0.0057}{0-100}\right|$ = **4.3 × 10^{-5} mol dm^{-3} s^{-1}**

From rate equation, **4.3 × 10^{-5} = k(0.500)(0.01)**

k = 8.6 × 10^{-3} mol^{-1} dm^{3} s^{-1}


[Br$_2$(aq)] /mol dm^{-3}
0.012
0.011
0.01
0.009
0.008
0.007
0.006
0.0057
0.005
0.004
0.003
0.002
0.001
0
$(t_{½})_1$ = 175 s
$(t_{½})_2$ = 180 s
time /s
0
50
100
150
175
200
250
300
350
355
400

2. (e)

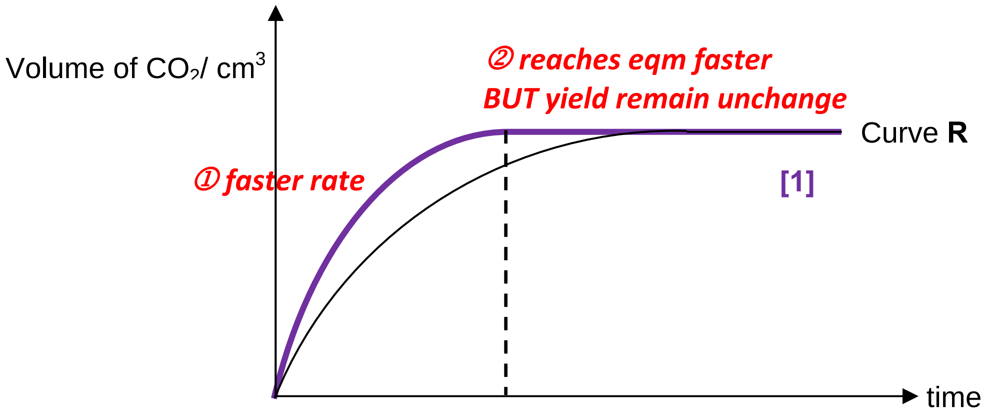
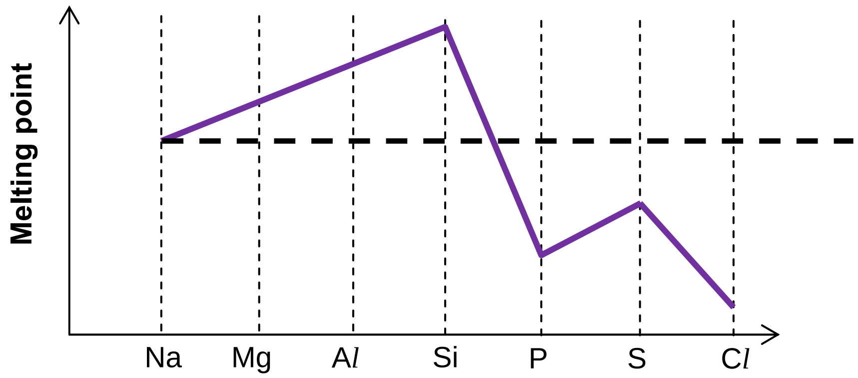
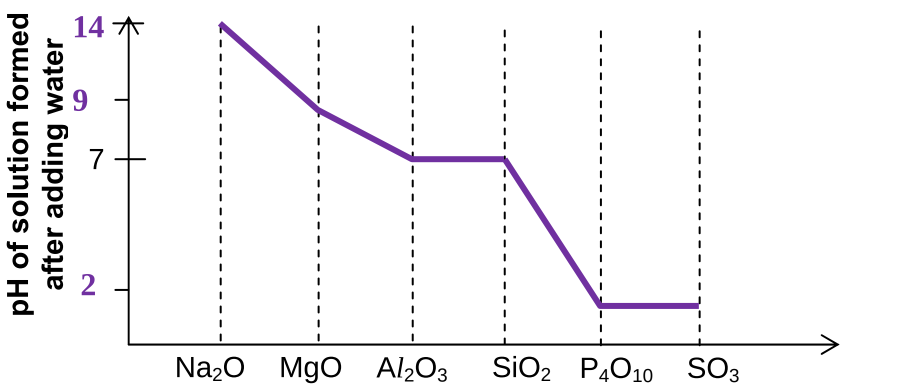


Figure 2.1

3. (a)





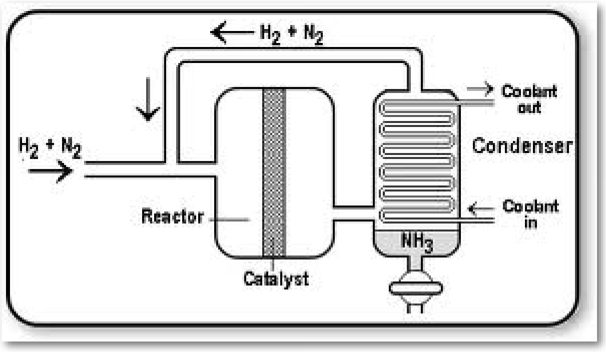
(b) **Mg^{2+} has a greater charge and a smaller radius than Na^+.** Moreover, **in Mg there are 2 valence electrons per Mg atom available for metallic bonding** while **in Na there is only one valence electron per Na atom available for metallic bonding**.

Thus **more energy is needed to overcome the stronger electrostatic attraction between Mg^{2+} and its valence electrons**.

(c) **$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $MgO + H_2O = Mg(OH)_2$**

Haber process is an industrial process to manufacture ammonia from nitrogen.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$



4. **(a)** Feature 1 : **rate of forward reaction = rate of backward reaction.**

Feature 2 : **Concentration of reactants and products remains constant** when a reaction is at equilibrium.

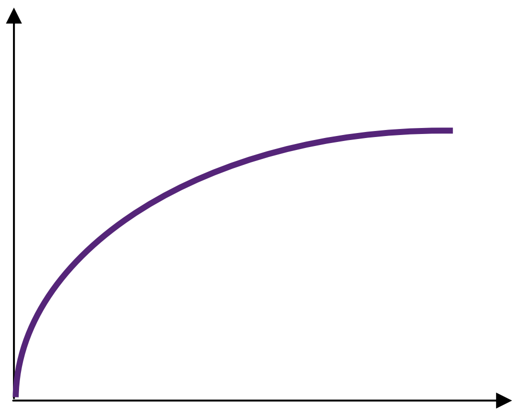
(b) **(i)** Temperature **450 °C**

Pressure **250 atm**

(ii) At **low temperature**, **equilibrium position shifts to the *right* to favour the exothermic reaction** and thus **increasing the yield** of ammonia.

However, **reaction is *slow* at low temperature**. Therefore, a ***moderately high*** temperature of about 450 °C is used.

(c) **(i)**



(ii) When pressure increase, **equilibrium position** of $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ **shifts to the right** to **decrease/reduce the number of gaseous molecules**.

Thus yield of NH_3 increases.

5. **(a)** Amount of $Na_2S_2O_3$ reacted in titration = $\frac{0.00100}{1000} \times 24.60 = 2.46 \times 10^{-5}$ mol

$2S_2O_3^{2-} \equiv I_2$

Amount of iodine reacted = $2.46 \times 10^{-5} \times \frac{1}{2} = 1.23 \times 10^{-5}$ mol

(b) **$2Mn(OH)_3 + 2I^- \rightarrow 2Mn(OH)_2(s) + 2OH^- + I_2$**

(c) $2Mn(OH)_3 \equiv I_2$

amount of $Mn(OH)_3$ reacted in **Step 2 = $1.23 \times 10^{-5} \times 2 = 2.46 \times 10^{-5}$ mol**

(d) Hence calculate the amount of $O_2(aq)$ in $25.0cm^3$ of the sample of river water.

$O_2 \equiv 4Mn(OH)_3$

amount of $O_2(aq)$ in $25.0cm^3$ of river water = **$2.46 \times 10^{-5} \div 4 = 6.15 \times 10^{-6}$ mol**

(e) **Concentration of O_2 in mol dm^{-3} =** $\frac{6.15 \times 10^{-6}}{25} \times 1000 = 2.46 \times 10^{-4}$ **mol dm^{-3}**

DOC = $2.46 \times 10^{-4} \times 2(16.0) = 0.00787$ g dm^{-3}

(f) **Since density of river water is 1 g cm^{-3},**

1000 g river water ----- 0.00787 g O_2(aq)

10^6 river water ------ $\frac{0.00787}{1000} \times 10^6 = 7.87$ **g of O_2(aq)**

DOC in ppm = 7.87

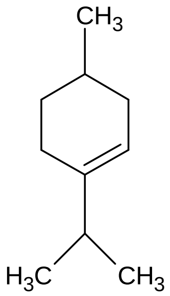
(f) **Yes** enough O_2 **because DOC > 5ppm.**

6. **(a)** **Alkene, tertiary alcohol**

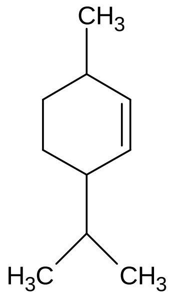
(b) **(i)** x = **3** y = **2**

(ii) **Excess concentrated H_2SO_4 heat (or 170°C)**

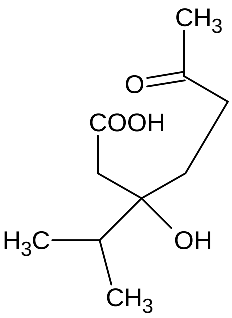
6. **(b)** **(iii)**



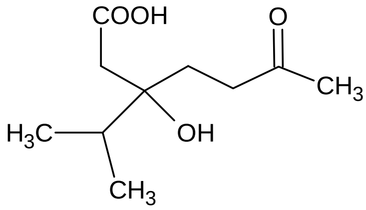
or



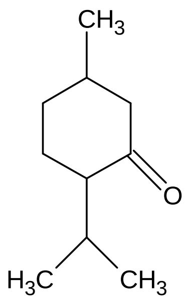
(c) **<u>Oxidation product</u>** of terpinen-4-ol:



or

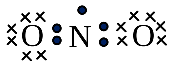


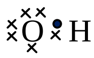
<u>Oxidation product</u> of menthol:



Section B

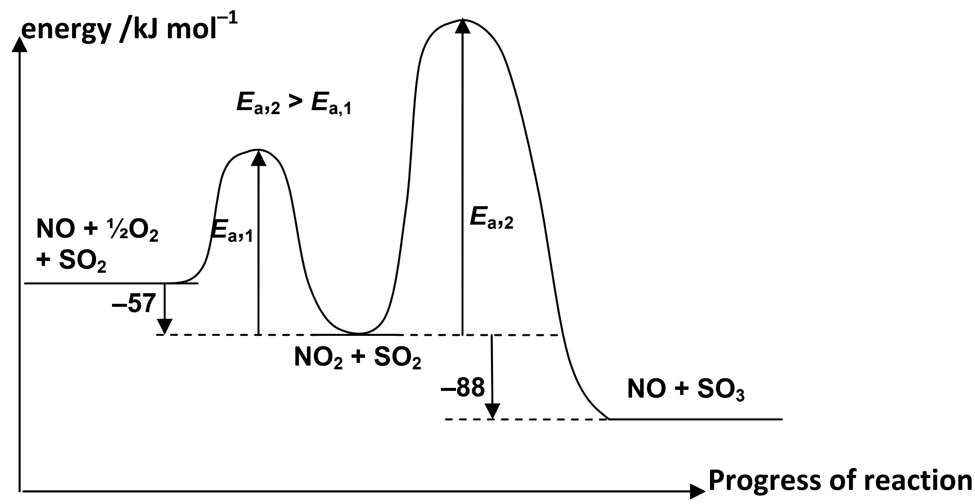
7. (a) (i)





React readily so that N and O can **<u>achieve octet configuration</u>** .

(ii)



(iii) **A catalyst <u>increases rate of reaction</u> by providing <u>an alternative reaction pathway with lowered activation energy</u>** It takes part in the reaction but remain chemically unchanged at the end of the reaction.

(b) (i) **$CS_2 + 2NO \rightarrow CO_2 + N_2 + 2S$**

Or

$4CS_2 + 8NO \rightarrow 4CO_2 + 4N_2 + S_8$

(ii) **Total volume of CO_2 and N_2 = 130 cm^3**

Volume of CO_2 (or N_2) = 65 cm^3

Amount of CO_2 (or N_2) = $\frac{65}{24000}$ = 0.00271 mol

$CS_2 \equiv N_2 \equiv CO_2$

Amount of CS_2 in sample = 0.00271 mol

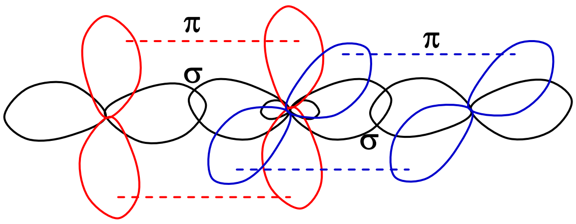
Mass of CS_2 in sample = 0.00271 $\times[2.0+2(32.1)]$ = 0.207 g

(iii) **$CO_2(g) + NaOH(aq) \rightarrow NaHCO_3(aq)$**

Or

$CO_2(g) + 2NaOH(aq) \rightarrow Na_2CO_3(aq) + H_2O(l)$

7. (b) (iv)



(c) (i) <u>**Excess NH_3 or limited CH_3CH_2X**</u>

(ii) **Since <u>C*l* has a smaller atomic radius</u> than I, <u>C-C*l* bond is shorter</u> and hence <u>stronger</u>. It requires more energy to break C-C*l* bond.**

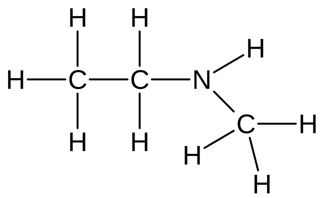
Thus <u>CH_3CH_2Cl is less reactive</u>.

(d) (i) Step I: <u>**ethanolic NaCN (or KCN), heat under reflux.**</u>

Step II: <u>**LiA*l*H_4 dry ether or H_2, Ni catalyst, heat**</u>

(ii) <u>**CH_3CH_2CN**</u>

(e)

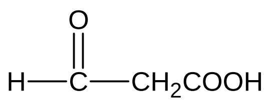


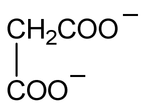
8. (a) (i) <u>**Process 1 uses a lower pressure which means lower cost or safer method.**</u>

Or

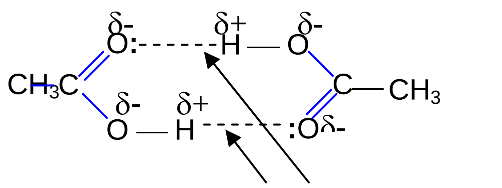
<u>**Process 1 produces high yield of ethanoic acid while Process 2 produces a large variety of products.**</u>

(ii)

L : 

Product of L : 

(b)



8. **(c)** **(i)** *Bronsted-Lowry acid* **is a proton donor.**

(ii) **$CH_3COOH < CH_3CH(OH)CO_2H < CH_3COCO_2H$**

(iii) **Since O is more electronegative than C, –CO group is electron withdrawing.**

Thus the negative charge on O atom of $CH_3COCO_2^-$ is more dispersed than that of $CH_3CO_2^-$.

Hence $CH_3COCO_2^-$ is more stable and CH_3COCO_2H is a stronger acid.

(iv)

	Reagents and Conditions	Type of reaction
Step **I**	**H_2, Ni catalyst, heat**	**Reduction**
Step **II**	**HC*l*, ZnCl_2 catalyst, heat**	**Substitution**

(d) **(i)** **Standard enthalpy change of neutralisation is the energy released when 1 mol of water is formed from an acid-base reaction under standard conditions.**

(ii) Ethanoic acid is a **weak acid** which partially dissociates.

Energy released during neutralisation is reabsorbed to **dissociate ethanoic acid completely**.

Thus standard enthalpy change of neutralisation of ethanoic acid is less exothermic.

(e) **(i)** **$K_a = 10^{-3.86}$ = $\underline{1.38 \times 10^{-4}}$ mol dm^{-3}**

$[H^+] = \sqrt{(0.25)(1.38\times10^{-4})} = 5.87 \times 10^{-3}$ mol dm^{-3}

pH = 2.23

(ii) **Amount of lactic acid used =** $\frac{25.0}{1000}\times0.25$ **= 6.25×10^{-3} mol**

Equivalence point is reached when 22 cm^3 NaOH is added.

Amount of NaOH in 22 cm^3 = 6.25×10^{-3} mol

Concentration of NaOH = $\frac{6.25\times10^{-3}}{22}\times1000$ **= 0.284 mol dm^{-3}**

8. (e) (iii) A buffer is a solution that resists pH changes when small amount of acid or base is added.

$$CH_3CH(OH)CO_2^- + H^+ \rightarrow CH_3CH(OH)CO_2H$$

(iv) Thymol blue

Colour change : yellow to green

9. **(a)** **P**, **Q** and **R** are three structural isomers with the molecular formula of $C_8H_8O_2$.

① **C: H = 1 : 1 ⇒ presence of benzene ring in P, Q and R**

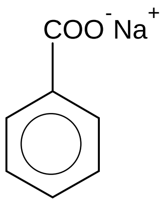
P and **Q** react with aqueous NaOH at room temperature.

② **Acid base reaction ⇒ P and Q contains –COOH**

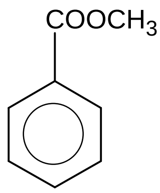
R does not react with aqueous NaOH at room temperature but when heated with aqueous NaOH, **R** forms compound **S**, $C_7H_5O_2Na$ and compound **T** which has a M_r of 32.

③ **Alkaline hydrolysis ⇒ R is an ester**

④**T is an alcohol.** **Since Mr of T= 32, T is CH_3OH**



S



R

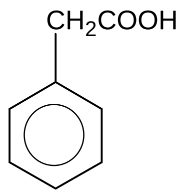
T forms white fumes with PCl_5.

⑤ ⇒ **substitution**

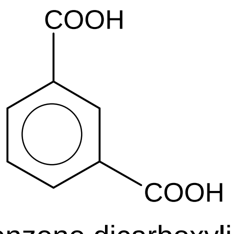
When heated with acidified $KMnO_4$, **P** and **R** oxidised to form the same product, benzoic acid.

⑥ ⇒ **P** and **R are monosubstituted benzene**

P is

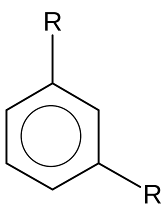


However, when heated with acidified $KMnO_4$, **Q** formed 1,3-benzene dicarboxylic acid.



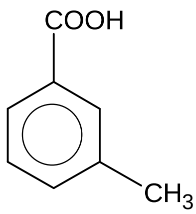
1,3-benzene dicarboxylic acid

Q has the structure

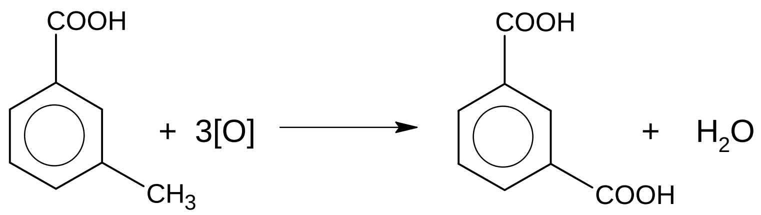


⑦

Q is



9. (a) (ii) $CH_3OH + PCl_5 \rightarrow CH_3Cl + POCl_3 + HCl$



(b) (i) **P can <u>expand octet</u> because it has <u>energetically accessible vacant 3d orbitals.</u> Hence it forms PCl_5.**

N <u>cannot expand octet</u> because its <u>next available vacant orbital is the 3s orbital</u>. <u>Promotion of electron to another principal quantum shell requires too much energy to be feasible.</u>

(ii) **Bond angle in PCl_3 is <u>107°</u>**.

In PCl_3, the **P** has **<u>3 bond pairs and 1 lone pair</u>**. The **<u>4 electron pairs will arrange in a tetrahedral arrangement</u>**.

However, since **<u>lone pair-bond pair repulsion is greater than the bond pair-bond pair repulsion</u>**, the bond angle is reduced from 109.5° to 107°.

(iii) **<u>PCl_5 has a larger number of electrons per molecule.</u>**

Thus **<u>more energy</u>** is needed to overcome the **<u>stronger van der Waals' forces between PCl_5 molecules than the weaker permanent dipole-permanent dipole interaction between the PCl_3 molecules</u>**.

As a result, PCl_5 has a higher melting point.

(c) (i) **<u>Student G</u> sees the <u>formation of white fumes</u>.**

<u>Student H</u> sees the <u>green solution turns red</u>.

(ii) $PCl_5(s) + H_2O(l) \rightarrow POCl_3(l) + 2HCl(g)$